



**BACHELOR OF COMPUTER SCIENCE
(SOFTWARE ENGINEERING) WITH HONOURS
SEMESTER II 2025/2026**

CSE3433

SOFTWARE ARCHITECTURE

GROUP 13: TALENTFLOW

**EVENT – DRIVEN JOB RECRUITMENT & APPLICATION
PROCESSING SYSTEM**

INDIVIDUAL LOGBOOK

GROUP MEMBERS	MATRIC NO.	ROLE
IZZ DHIA DAMIA BINTI MAT ZAIDI	S75849	BACKEND & INTEGRATION LEAD
NURUL IFFAH SABRINA BINTI AHMAD SHUKRI	S75894	SYSTEM ARCHITEC & DESIGNER

WEEKLYP PROGRESS

PHASE 1 (WEEK 3 – WEEK 5) – PROJECT PROPOSAL

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 3	Initial system idea & concept formation	Defined TalentFlow concept and discussed event-driven recruitment system ideas	Unclear understanding of Event-Driven Architecture (EDA) and how it applies to recruitment systems	Conduct deeper research on EDA principles, publish-subscribe models, and real-world event-driven recruitment platforms. Study how RabbitMQ and Kafka work as message brokers. Break down the recruitment process into event-driven workflows. Use AI tools and references to validate concept clarity
Week 4	Requirement analysis & project validation	Refined system features and explored event-driven architecture implementation	Difficulty understanding asynchronous communication and event flow design	Study EDA case studies and identify how events trigger actions in recruitment systems. Separate functional and non-functional requirements clearly. Validate system direction with teammates and refine scope based on lecturer expectations
Week 5	Proposal drafting & finalisation	Defined functional and non-functional requirements, selected EDA as architecture style	Time pressure before submission deadline and ensuring all EDA concepts are correctly explained	Review entire proposal structure including problem description, objectives, system requirements, and architecture justification. Perform proofreading, formatting correction, and final validation before submission

PHASE 2 (WEEK 6 – WEEK 7) – ARCHITECTURE DESIGN

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 6	Study Event-Driven Architecture and publish-subscribe model	Researched EDA concepts, message brokers, and system decomposition	Limited understanding of how to decompose a recruitment system into event-driven services	Strengthen conceptual understanding by studying EDA examples and breaking the recruitment process into event producers, consumers, and event channels. Map TalentFlow system into services such as Job Service, Application Service, Resume Filter, Status Management, Interview Service, Notification Service, and Dashboard Service
Week 7	Architecture design and system diagram preparation	Designed component architecture, event flow, and created UML diagrams	Complexity in designing event flow diagrams and ensuring loose coupling between services	Improve diagram structure by ensuring proper relationships between system components and event flows. Validate architecture with system requirements and ensure consistency between architecture diagrams, event flow, and system workflow. Refine based on feedback from team discussion

PHASE 3 (WEEK 8 – WEEK 11) – PROTOTYPE PROGRESS

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 8	Prototype development planning and module scope definition	Defined core event flow for prototype implementation focusing on application submission to notification flow	Determining the appropriate scope for the prototype and which events to implement first	Focus on building the core event flow: Candidate Submits Application → Application Service → Resume Filter Service → Status Management → Notification Service. Plan RabbitMQ setup and service communication
Week 9	System evaluation & debugging of event-driven components	Tested message broker connection and event publishing/consuming	RabbitMQ connection issues and event delivery failures	Debug backend systematically by isolating each service and testing event publishing and consuming individually. Ensure proper queue binding and message acknowledgment. Document event flow for integration consistency
Week 10	Backend development, message broker integration and deployment preparation	Developed core services using Node.js/Express, integrated RabbitMQ, and explored deployment setup	First-time experience with RabbitMQ and message broker configuration	Follow RabbitMQ documentation step-by-step for installation and configuration. Implement publisher and consumer patterns. Test event flow thoroughly before deployment. Identify configuration errors and refine process
Week 11	System enhancement, testing and cloud deployment	Improved event handling, fixed bugs and deployed prototype on Render	Deployment configuration issues and ensuring message broker works in cloud environment	Perform full system testing after deployment to ensure event flow works correctly. Monitor cloud deployment performance and continuously troubleshoot issues. Test all event-driven workflows to ensure system reliability

PHASE 4 (WEEK 12 – WEEK 13) – PROJECT REPORT, INDIVIDUAL LOGBOOK & FINAL DEMO

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 12	Final system review and report preparation	Verified deployed system functionality, tested all event flows, and reviewed documentation	Documentation structure and ensuring all EDA concepts are properly explained	Validate system against all requirements and testing checklist. Ensure all event flows function correctly. Follow official report template strictly and reorganise content for clarity, consistency, and academic formatting standards
Week 13	Final report submission, presentation, and system demo	Completed project report and individual logbook, prepared presentation slides, demo script, and conducted system demonstration	Formatting inconsistencies in documentation and nervousness for presentation	Standardise formatting across all documents, perform final proofreading, conduct multiple rehearsals to improve confidence and presentation flow, and prepare clear technical explanations and answers for the presentation session

GENAI USAGE RECORD

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	01/04/2026	DeepSeek	Project concept	“what is event-driven architecture, explain dengan simple”	Explanation about event-driven architecture	Used to choose group project title and to explain in proposal report
			System exploration	“list system untuk event- driven job recruitment and application processing system”	Short explanation about event-driven architecture and example of that system	Used to compare and choose project title and ideas
Week 4	10/04/2026	DeepSeek	System analysis	“explain tentang TalentFlow and show main components in that system”	Details explanation about TalentFlow and list of main components	In proposal report
Week 5	16/04/2026	DeepSeek	System requirement	“initial functional requirement yang ada dalam TalentFlow in event-driven architecture”	List of initial functional requirement with explanation	In proposal report
Week 7	28/04/2025	DeepSeek	Architecture diagram	“show me event-driven architecture diagram for job requirement & application processing system”	Example for architecture diagram as system context diagram, component/service architecture diagram, interaction diagram and deployment diagram	Used in architecture design report
			Architecture explanation	“we choose EDA, can you explain why use EDA for recruitment”	Explanation about EDA and Broker topology	Used in architecture design report
			Architecture understanding	“it is correct for component/service architecture diagram (i send picture to check)”	Corrected the diagram by telling which part is wrong	Used in architecture design report
Week 12	09/06/2026	DeepSeek	Report understanding	“contoh non functional yang ada dalam EDA talentFlow”	Provided list of functional requirement with short explanation	Used in final project report

				“ape techonology stack dalam TalentFlow”	List of architecture layer with technology used and purpose	Final project report
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